

## Unit 8, Lesson 10: Solving Systems of Equations by Graphing for Real-World Contexts

### Benchmark

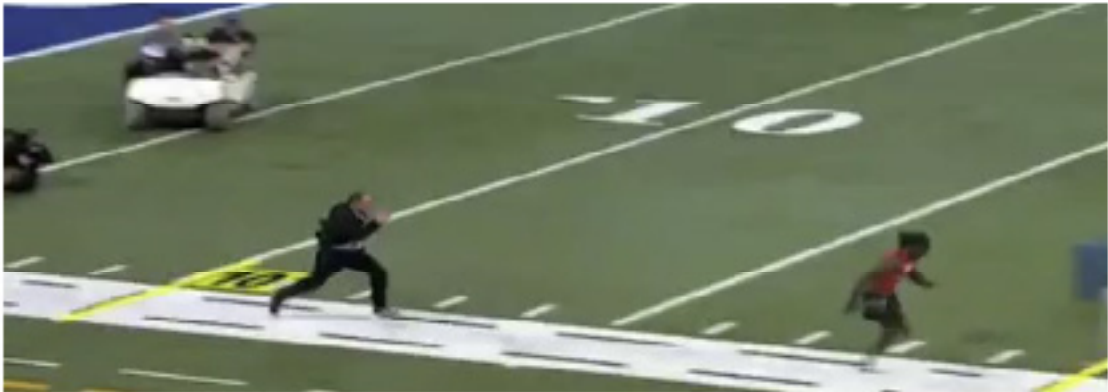
**MA.8.AR.4.3** Given a mathematical or real-world context, solve systems of two linear equations by graphing.

### Learning Target

- ☐ I can solve a system of two linear equations by graphing given a real-world context.

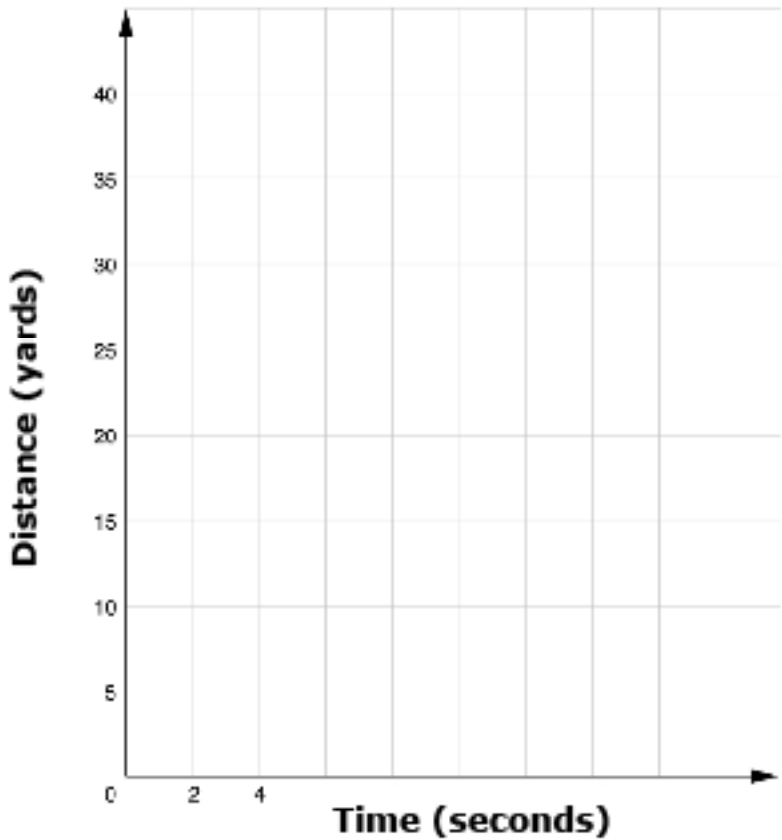
8.10.1 Warm-Up: Notice and Wonder

1. Julio Jones was drafted to the Atlanta Falcons in 2011. The same year, a video was created showing Rich Eisen, an NFL correspondent, side-by-side with Julio Jones running a 40-yard dash. A screenshot from the video is shown.



a. What do you notice?

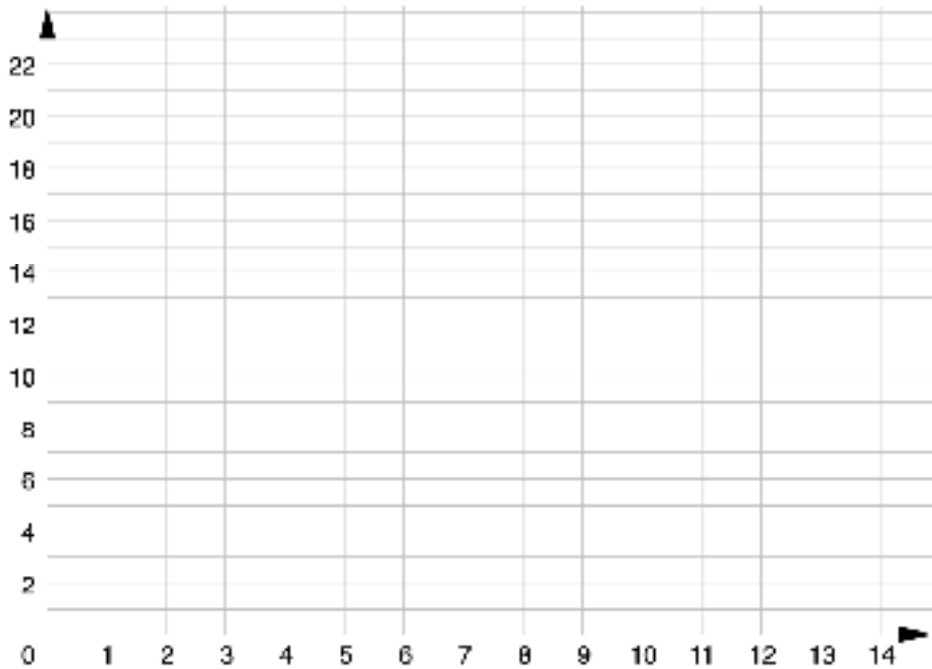
b. Make a sketch of what you think the graph of the two runners would look like. Be sure to label each line with the name of the runner.



8.10.2 Guided Instruction: Solving Systems of Equations by Graphing in Real-World Contexts

Bamboo plants are known for growing very quickly. Imani measures the height of two bamboo plants over the course of several days. Plant A starts at 6 feet (ft) tall and grows at a constant rate of  $\frac{1}{4}$  ft each day. Plant B starts at 3 ft tall and grows at a constant rate of  $\frac{1}{2}$  ft each day.

1. The system  $\begin{cases} y = \frac{1}{4}x + 6 \\ y = \frac{1}{2}x + 3 \end{cases}$  can be used to represent this situation, where  $x$  is the number of days and  $y$  is the height of the bamboo plant.
- a. Graph the system of equations.

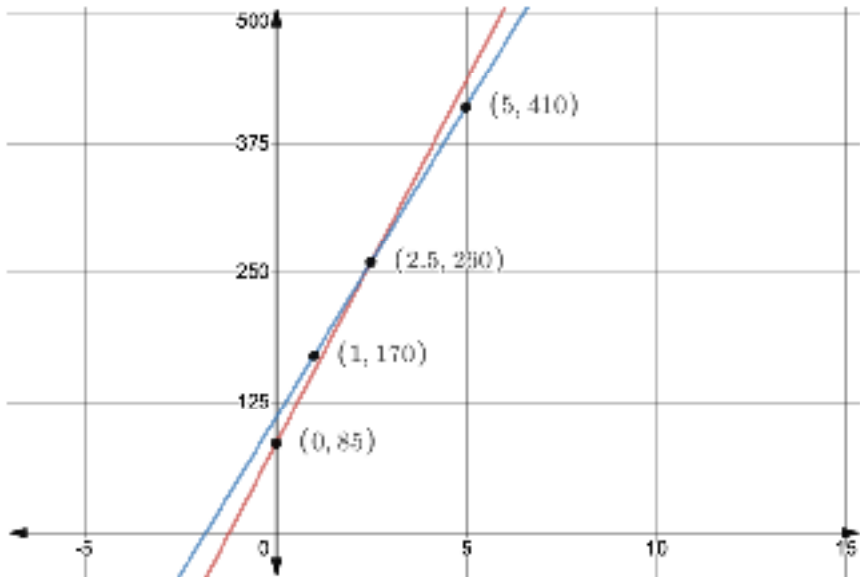


- b. Label the axes with appropriate titles and units.
- c. Label each line Plant A or Plant B.
- d. Mark the solution to the system of equations on the graph.

- e. What are the coordinates of the solution?
- f. Explain the meaning of the solution in this context.
2. Sofia is deciding between two internet service providers. The first plan costs \$70 per month, with a one-time charge of \$85 for equipment. The second company charges \$60 per month, but the one-time equipment charge is \$110.

The system  $\begin{cases} y = 70x + 85 \\ y = 60x + 110 \end{cases}$  represents this situation, where  $x$  is the number of months and  $y$  is the total cost.

The graph of the system is shown with the coordinates of four points labeled.



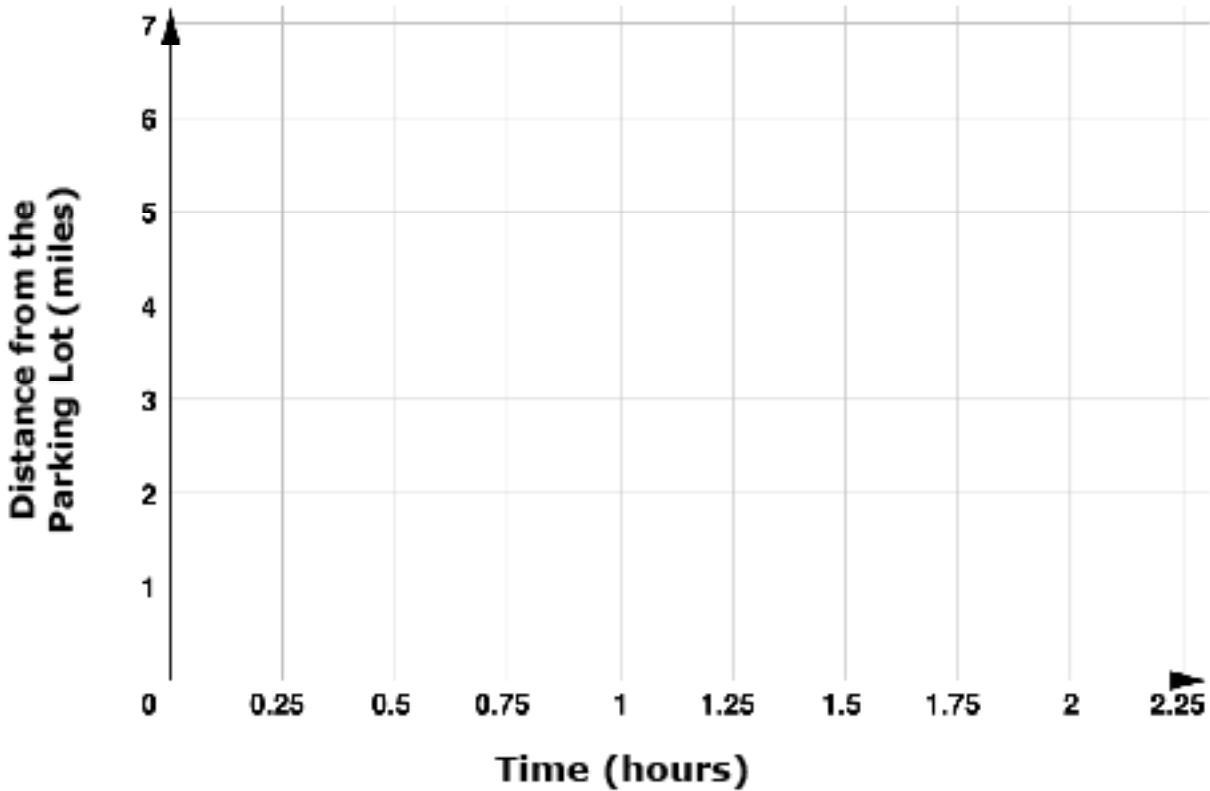
- a. What is the solution to the system?
- b. Explain the meaning of the solution in this context.

8.10.3 Guided Practice: Solving Systems of Equations by Graphing for Real-World Contexts

1. Chauncey and Anais both decide to head to the park. They hike from the parking lot to the lake and back, but they start their hikes at different times.

At the time that Chauncey reaches the lake and starts to turn back, Anais is 1 mile away from the parking lot and hiking at a constant speed of 3.75 miles per hour toward the lake. Anais’ distance from the parking lot,  $d$ , can be expressed as  $d = 3.75t + 1$ , where  $t$  is the number of hours since Chauncey started walking back toward the parking lot.

Chauncey’s distance from the parking lot,  $d$ , can be expressed as  $d = -2.25t + 5$ , where  $t$  represents the time, in hours, since he left the lake.



- a. Graph both lines on the coordinate plane.
- b. Label each line as “Anais” or “Chauncey.”

- c. What is the approximate solution to the system  $\begin{cases} d = 3.75t + 1 \\ d = -2.25t + 5 \end{cases}$ ?
- d. Explain the meaning of the solution in context.
- e. Explain why only the approximate solution for the system can be determined by graphing.

8.10.4 Collaboration: Graphing Real-World Systems

1. Victor is monitoring the water levels in the control room at a water treatment plant. He receives an alert that Tank A has a leak. It started with 1,500 liters of water and is losing 2 liters of water per minute. At the same time as the leak began, Tank B held 300 liters of water and is being filled at a rate of 6 liters per minute.

This situation can be represented by the system  $\begin{cases} y = -2x + 1500 \\ y = 6x + 300 \end{cases}$ , where  $x$  is the amount of time, in minutes, since Victor received the alert and  $y$  is the amount of water, in liters, in the tank.

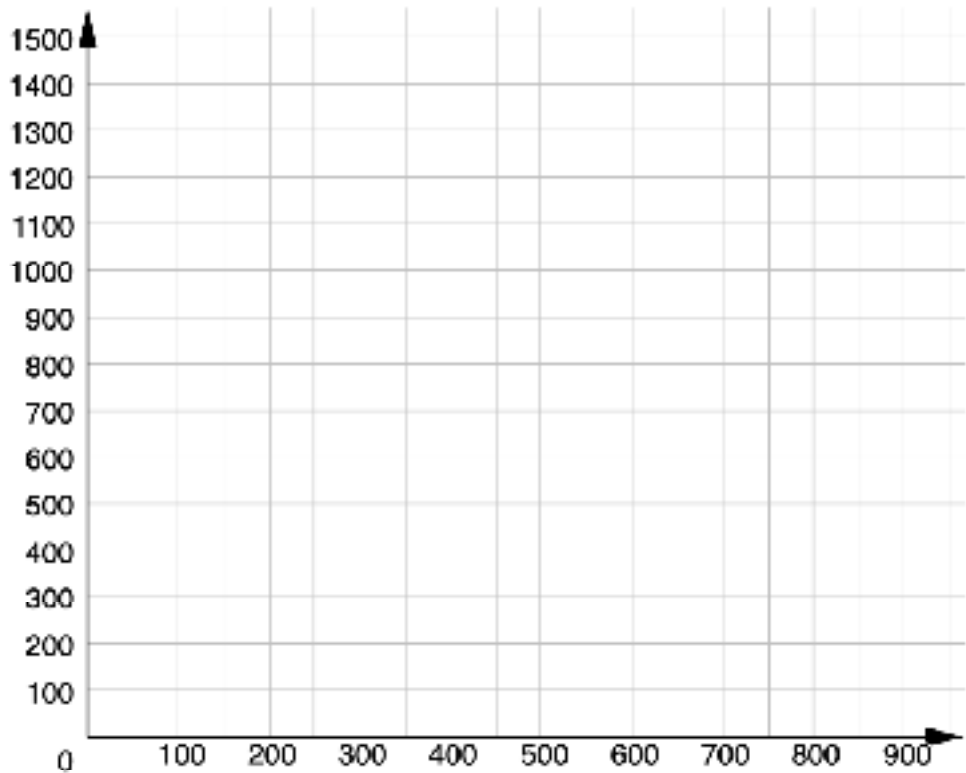
- a. Discuss with your partner which equation represents each tank. Then, complete the table.

| Tank | Equation         |
|------|------------------|
|      | $y = -2x + 1500$ |
|      | $y = 6x + 300$   |

- b. Work with your partner to determine how much water will be in each tank 100 minutes after Victor received the alert.

| Tank | Volume, in Liters, After 100 Minutes |
|------|--------------------------------------|
| A    |                                      |
| B    |                                      |

c. On your own, graph the system of equations on the coordinate grid.



- d. Label both axes with appropriate titles and units.
- e. Compare your graph and labels with your partner and resolve any differences, if necessary.
- f. What is the solution to the system?
- g. Explain what the solution means in this context.

8.10.5 Wrap-Up: Reflection

1. Take a SELFIE by checking the box that best matches what you did today.

|                                  |                          |                          |
|----------------------------------|--------------------------|--------------------------|
| Showed my work                   | <input type="checkbox"/> | <input type="checkbox"/> |
| Explained my answers             | <input type="checkbox"/> | <input type="checkbox"/> |
| Lots of math vocabulary used     | <input type="checkbox"/> | <input type="checkbox"/> |
| Found solutions by graphing      | <input type="checkbox"/> | <input type="checkbox"/> |
| I persevered through the problem | <input type="checkbox"/> | <input type="checkbox"/> |
| Eliminated careless errors       | <input type="checkbox"/> | <input type="checkbox"/> |

